Covariate-matched null-hypothesis ranges with nullranges: :**matchRanges( )**

Docs: https://nullranges.github.io/nullranges/ Code: https://github.com/nullranges/nullranges/

Basics

When performing statistical analysis on a set of genomic regions with different features (e.g. region length, signal strength), it is often important to compare to null sets that preserve these potential covariate features.

To address this need, the nullranges package implements matchRanges(), an efficient and convenient tool for selecting a subset of covariate-matched null-hypothesis ranges from a pool of background ranges.

The package provides a host of functions for assessing, visualizing, and extracting matched data that integrate seamlessly into existing Bioconductor workflows.

Usage

```
matchRanges(focal,
            pool,
            covar = ~feat1 + feat2,
            method = 'rejection',
            replace = FALSE)
```

*Ranges of interest

*Ranges from which to select matches

Features to match

Matching method

With or without replacement

*Also accepts paired ranges (i.e. GInteractions) and DataFrames/data.tables.

Developers

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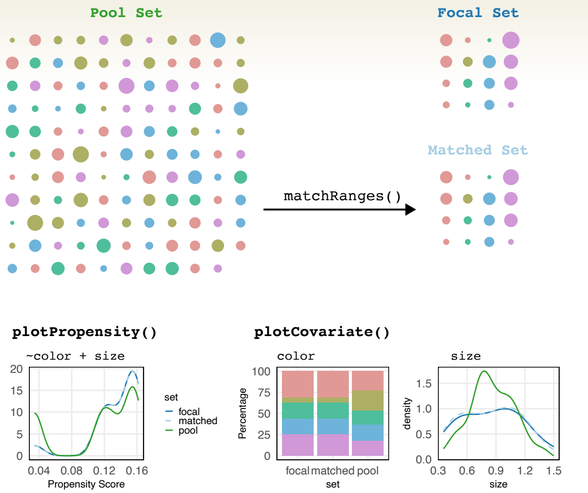
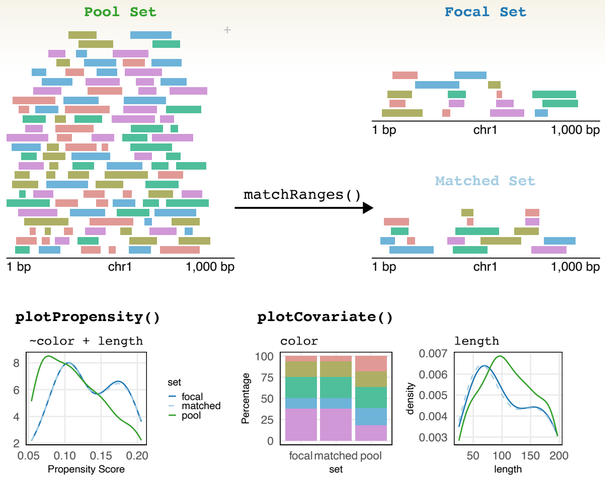
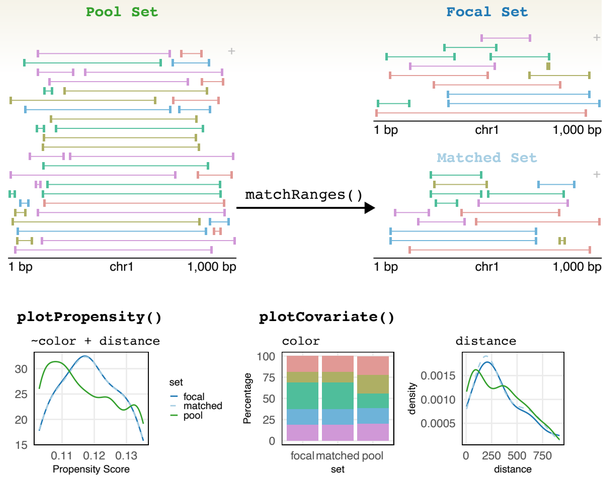
Matching with DataFrames



Matching with Ranges



Matching with Paired Ranges



Assessing Quality

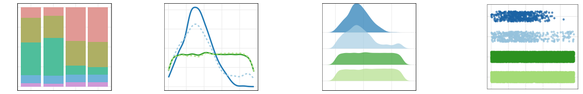
```
x <- matchRanges(focal, pool, covar, ...)
```

```
overview(x, digits)
```

```
plotPropensity(x, sets, type, ...) &
plotCovariate(x, covar, sets, type, ...)
```

Discrete: type='bars'

Continuous: type='lines', type='ridges', type='jitter'



Accessor Functions

Get data from matched objects

matchedData(x)
A data.table with matched data

covariates(x)
A character vector of covariates

method(x)
Matching method used in x

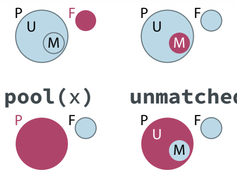
withReplacement(x)
Matching with or without replacement

indices(x, set)
Indices for a given set

Sets

Get sets from matched objects

focal(x) **matched(x)** **pool(x)** **unmatched(x)**



*Returns objects of the same class

Case Studies

In-depth examples applying matchRanges to biological data

Case Study I: CTCF Occupancy

Here, we generate a set of null-hypothesis GRanges to test for the occupancy of CCCTC-binding factor (CTCF) at chromatin loop anchors independently from open chromatin regions.

Case Study II: CTCF Orientation

The convergence rule purports that most CTCF-bound loops occur in a convergent orientation, where the asymetric CTCF motifs at each anchor face one another. Here, we generate a set of null-hypothesis paired ranges to test the convergence rule independently of CTCF strength, number of sites, and pair distance.

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